

Why vLLM is fast

vLLM's two ideas

PagedAttention: the KV cache is sliced into fixed-size BLOCKS, allocated on demand, like OS virtual-memory pages.

continuous batching: a finished request's slot is handed to the next queued request immediately, not at the next batch boundary.

Both ideas raise GPU utilization by shrinking wasted memory and wasted idle time, not by changing the math.